

Chapter One

LORENTZ

TRANSFORMATION

1.3 Relative Velocity

Let the co-ordinates of a particle in frame S be (x, y, z, t) and in frame $S'(x', y', z', t')$ then the component of its velocity in two frame can be written as

There is one inertial frame S , S' is another inertial frame moving with respect to S in x -direction and A is another inertial frame which is moving with respect to S' with velocity component (u'_x, u'_y, u'_z) .

$$\text{So, } u'_x = \frac{dx'}{dt'}, u'_y = \frac{dy'}{dt'}, u'_z = \frac{dz'}{dt'}$$

The velocity component of A from S frame is given by

$$u_x = \frac{dx}{dt}, u_y = \frac{dy}{dt}, u_z = \frac{dz}{dt}$$

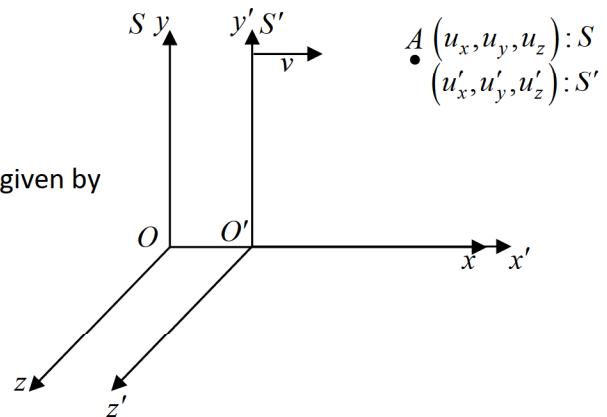


Figure: 1.4

From the inverse Lorentz Transformation,

$$x = \gamma(x' + vt') \text{ , } y = y' \text{ , } z = z' \text{ , } t = \gamma\left(t' + \frac{vx'}{c^2}\right) \text{ where } \gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

differentiating both sides, we get

$$dx = \gamma(dx' + vdt') \text{ , } dy = dy' \text{ , } dz = dz' \text{ and } dt = \gamma\left(dt' + \frac{vdx'}{c^2}\right)$$

$$u_x = \frac{dx}{dt} = \frac{\gamma}{\gamma} \frac{\frac{dx'}{dt'} + v}{1 + \frac{v}{c^2} \frac{dx'}{dt'}} = \frac{\frac{dx'}{dt'} + v}{1 + \frac{v}{c^2} \frac{dx'}{dt'}} \Rightarrow u_x = \frac{u'_x + v}{1 + \frac{v}{c^2} u'_x}$$

$$u_y = \frac{dy}{dt} = \frac{dy'}{\gamma\left(dt' + \frac{vdx'}{c^2}\right)} = \frac{\frac{dy'}{dt'}}{\gamma\left(1 + \frac{v}{c^2} \frac{dx'}{dt'}\right)} \Rightarrow u_y = \frac{u'_y \sqrt{1 - \frac{v^2}{c^2}}}{\left(1 + \frac{vu'_x}{c^2}\right)}$$

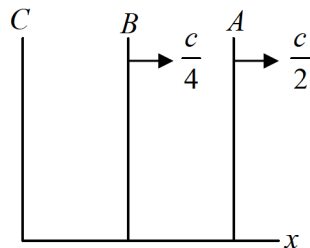
$$u_z = \frac{dz}{dt} = \frac{dz'}{\gamma\left(dt' + \frac{vdx'}{c^2}\right)} = \frac{u'_z \sqrt{1 - \frac{v^2}{c^2}}}{\left(1 + \frac{v}{c^2} \frac{dx'}{dt'}\right)} \Rightarrow u_z = \frac{u'_z \sqrt{1 - \frac{v^2}{c^2}}}{\left(1 + \frac{vu'_x}{c^2}\right)}$$

From Lorentz transformation we can get $u'_x = \frac{u_x - v}{1 - \frac{vu_x}{c^2}}$, $u'_y = \frac{u_y \sqrt{1 - \frac{v^2}{c^2}}}{\left(1 - \frac{vu_x}{c^2}\right)}$, $u'_z = \frac{u_z \sqrt{1 - \frac{v^2}{c^2}}}{\left(1 - \frac{vu_x}{c^2}\right)}$

Example: A frame A move with respect to frame B with speed $\frac{c}{2}$ in positive $\hat{x}$

direction .Another frame C moves with respect to B with speed $\frac{c}{4}$ in positive $\hat{x}$ with speed $\frac{c}{4}$.

What will speed of C with respect to observer attached to frame A ?



Solution: It is given $v_{B,A} = \frac{c}{2}$ and $v_{C,B} = \frac{c}{4}$

We assume frame A is equivalent to S then frame B as S'

The given information is $v_{B,A} = v_{S',S} = v = \frac{c}{2}\hat{i}$ $v_{C,B} = v_{C,S'} = u'_x = \frac{c}{4}$ and $v_{C,A} = u_x = ?$

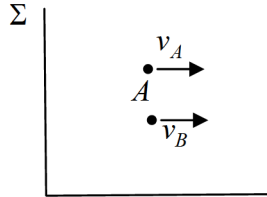
So we use formula $u_x = \frac{u'_x + v}{1 + \frac{u'_x v}{c^2}} = \frac{\frac{c}{4} + \frac{c}{2}}{1 + \frac{\frac{c}{4} \cdot \frac{c}{2}}{c^2}} = \frac{\frac{c+2c}{4}}{1 + \frac{1}{8}} = \frac{\frac{3c}{4}}{\frac{9}{8}} = \frac{2}{3}c$

Example: Two particles A, B moving with respect to common frame Σ . If speed of A, B is measured by observer attached to Σ are v_A and v_B in positive $\hat{x}$ direction respectively.

(a) By using Lorentz transformation Find the speed of B with respect to observer attached to A .

(b) Find the speed of B with respect to observer attached to A if $\frac{v_B}{c} \rightarrow 0$

(c) Find the speed of B with respect to observer attached to A if $v_B = c$



Solution:

(a) $v_{A,\Sigma} = v_A \hat{i}$ $v_{B,\Sigma} = v_B \hat{i}$ $v_{B,A} = ?$

Assume particle A as S so Σ is identified as S'

So $v_{\Sigma,A} = v_{S',S} = v = -v_A \hat{i}$, $v_{B,\Sigma} = v_{B,S'} = u'_x = v_B \hat{i}$ and $v_{B,A} = v_{B,S} = u_x = ?$

$$v_{B,A} = u_x = \frac{u'_x + v}{1 + \frac{u'_x v}{c^2}} = \frac{v_B + (-v_A)}{1 + \frac{v_B \cdot (-v_A)}{c^2}} = \frac{v_B - v_A}{1 - \frac{v_B \cdot v_A}{c^2}}$$

(b) $v_{B,A} = u_x = \frac{v_B - v_A}{1 - \frac{v_B \cdot v_A}{c^2}}$ if $\frac{v_B}{c} \rightarrow 0$ $u_x = v_B - v_A$, which is exactly result obtain by Galilean

transformation

$$(c) v_{B,A} = u_x = \frac{v_B - v_A}{1 - \frac{v_B \cdot v_A}{c^2}} \quad v_B = c$$

$$v_{B,A} = u_x = \frac{c - v_A}{1 - \frac{c \cdot v_A}{c^2}} = \left(\frac{c - v_A}{c - v_A} \right) c = c, \text{ which endorsed the postulates that speed of light is}$$

invariant from inertial frame of reference.

Example: Two particles A, B moving with respect to common inertial frame Σ . If speed of A, B is measured by observer attached to Σ are v_A and v_B in positive $\hat{x}$ and $\hat{y}$ direction respectively.

(a) By using Lorentz transformation Find the velocity of B with respect to observer attached to A .

(b) Find the speed of B with respect to observer attached to A if $v_B = c$

$$v_{A,\Sigma} = v_A \hat{i} \quad v_{B,\Sigma} = v_B \hat{j} \quad v_{B,A} = ?$$

Solution:

(a) Assume particle A as S so Σ is identified as S'

So $v_{\Sigma,A} = v_{S',S} = v = -v_A \hat{i}$, $\vec{v}_{B,\Sigma} = \vec{v}_{B,S'} = u'_x = 0, u'_y = v_B, u'_z = 0$ and $\vec{v}_{B,A} = \vec{v}_{B,S} = u_x \hat{i} + u_y \hat{j} + u_z \hat{k} = ?$

$$u_x = \frac{u'_x + v}{1 + \frac{u'_x v}{c^2}} = \frac{0 + (-v_A)}{1 + \frac{0 \cdot (-v_A)}{c^2}} = \frac{0 - v_A}{1 - \frac{0 v_A}{c^2}} = -v_A, \quad u_y = \frac{u'_y \sqrt{1 - \frac{v^2}{c^2}}}{1 + \frac{u'_x v}{c^2}} = \frac{v_B \sqrt{1 - \frac{(-v_A)^2}{c^2}}}{1 + \frac{0 \cdot (-v_A)}{c^2}} = v_B \sqrt{1 - \frac{v_A^2}{c^2}}$$

$$\text{Similarly, } u_z = \frac{u'_z \sqrt{1 - \frac{v^2}{c^2}}}{1 + \frac{u'_x v}{c^2}} = \frac{0 \sqrt{1 - \frac{(-v_A)^2}{c^2}}}{1 + \frac{0 \cdot (-v_A)}{c^2}} = 0, \quad \vec{v}_{B,A} = \vec{v}_{B,S} = u_x \hat{i} + u_y \hat{j} + u_z \hat{k} = -v_A \hat{i} + v_B \sqrt{1 - \frac{v_A^2}{c^2}} \hat{j}$$

$$(b) \text{ For } v_B = c \quad \vec{v}_{B,A} = -v_A \hat{i} + c \sqrt{1 - \frac{v_A^2}{c^2}} \hat{j}$$

$$|v_{B,A}| = \sqrt{u_x^2 + u_y^2 + u_z^2} = \sqrt{v_A^2 + c^2 \left(1 - \frac{v_A^2}{c^2} \right)} = c, \text{ again we have seen the speed of light is invariant}$$

from any inertial frame.

Example: A spaceship moving away from the earth with velocity $0.5c$ fires a rocket whose velocity relative to space is $0.5c$ (a) Away from earth (b) Towards the earth. Calculate velocity of the rocket as observed from the earth in two cases.

Solution: (a) Assume that spaceship moving with respect to Earth in positive $\hat{x}$ direction.

So velocity of space ship with respect to Earth is $v_{SP,E} = 0.5$, $v_{R,SP} = 0.5c$ away from earth $v_{R,E} = ?$

Assume Earth as S frame, then space ship is S' frame and speed of Rocket is given with respect to spaceship i.e. S' .

$$v_{SP,E} = v_{S',S} = v = 0.5c, v_{R,SP} = v_{R,S'} = u'_x = 0.5c$$

$$(a) u_x = \frac{u'_x + v}{1 + \frac{u'_x v}{c^2}} = \frac{0.5c + 0.5c}{1 + \frac{0.5c \times 0.5c}{c^2}} = \frac{c}{1 + 0.25} = \frac{c}{1.25} = \frac{4}{5}c \Rightarrow u = 0.8c$$

$$(b) v_{SP,E} = v_{S',S} = v = 0.5c, v_{R,SP} = v_{R,S'} = u'_x = -0.5c \text{ rocket is moving towards earth } v_{R,E} = ?$$

$$u_x = \frac{u'_x + v}{1 + \frac{u'_x v}{c^2}} = \frac{-0.5c + 0.5c}{1 + \frac{-0.5c \times 0.5c}{c^2}} = 0$$

Example: A glass plate of refractive index μ moving with speed v in positive $\hat{x}$ - direction with respect to laboratory. Light pulse is entered into glass plate along $\hat{x}$ - axis.

(a) Find the speed of light in the glass plate with respect to observer attached to Glass plate.

(b) Find the speed of light in the glass plate with respect to observer attached to Laboratory

Solution: (a) If observer is attached to glass plate then glass plate is rest with respect to observer then speed of light in the glass plate is $v_{l,g} = \frac{c}{\mu}$.

$$(b) \text{ If observer is attached to laboratory then } v_{g,L} = v\hat{x}, v_{l,g} = \frac{c}{\mu}$$

We assume laboratory as S frame then glass plate is treated as S' so $v_{g,L} = v_{S,S'} = v\hat{x}$,

$$\text{so } v_{l,g} = v_{l,S'} = u'_x = \frac{c}{\mu}, v_{l,L} = v_{l,S} = u_x = ?$$

$$u_x = \frac{u'_x + v}{1 + \frac{u'_x v}{c^2}} = \frac{\frac{c}{\mu} + v}{1 + \frac{\frac{c}{\mu} \times v}{c^2}} = \left(\frac{c + \mu v}{\mu c + v} \right) c$$

Example: A light pulse emerge from origin of frame A . If observer is attached to frame the light ray is appeared to make angle θ_0 with x axis and movement of light is confine in $x-y$ plane. Frame A is moving with respect to frame B with speed v in positive x -axis. If observer will be attached with frame B , what will velocity of light pulse?

Solution: Assume frame B is S frame, Then frame A is treated as S' so $v_{S',S} = v \hat{x}$

Speed of light with respect to frame A in $\hat{x}$ direction is identified as $u'_x = c \cos \theta_0$

Speed of light with respect to frame A in $\hat{y}$ direction is identified as $u'_y = c \sin \theta_0$

Speed of light with respect to frame A in $\hat{z}$ direction is identified as $u'_z = 0$

We need to find velocity of light with respect to S frame i.e. $u_x = ?$, $u_y = ?$, $u_z = ?$

$$u_x = \frac{u'_x + v}{1 + \frac{u'_x v}{c^2}} = \frac{c \cos \theta_0 + v}{1 + \frac{c \cos \theta_0 v}{c^2}} = \left(\frac{c \cos \theta_0 + v}{c + v \cos \theta_0} \right) c$$

$$u_y = \frac{u'_y \sqrt{1 - \frac{v^2}{c^2}}}{1 + \frac{u'_x v}{c^2}} = \frac{c \sin \theta_0 \sqrt{1 - \frac{v^2}{c^2}}}{1 + \frac{c \cos \theta_0 v}{c^2}} = \left(\frac{c \sin \theta_0 \sqrt{1 - \frac{v^2}{c^2}}}{c + v \cos \theta_0} \right) c$$

$$u_z = \frac{u'_z \sqrt{1 - \frac{v^2}{c^2}}}{1 + \frac{u'_x v}{c^2}} = \frac{0 \times \sqrt{1 - \frac{v^2}{c^2}}}{1 + \frac{c \cos \theta_0 v}{c^2}} = 0$$

$$\vec{v}_{l,B} = \vec{u} = u_x \hat{i} + u_y \hat{j} + u_z \hat{k} = \left(\frac{c \cos \theta_0 + v}{c + v \cos \theta_0} \right) c \hat{i} + \left(\frac{c \sin \theta_0 \sqrt{1 - \frac{v^2}{c^2}}}{c + v \cos \theta_0} \right) c \hat{j}$$

